1 Case $n = 1$: The Midpoint Foundation

Goal: Find c_1 and x_1 such that $\int_{-1}^{1} f(x)dx \approx c_1 f(x_1)$ is exact for $f(x) = a_0 + a_1 x$.

Since we have 2 unknowns, we test the formula against the first two power functions:

1. **Test $f(x) = 1$ (Constant):**
   - Exact: $\int_{-1}^{1} 1dx = [x]_{-1}^{1} = 1 - (-1) = 2$.
   - Approximation: $c_1 \cdot f(x_1) = c_1 \cdot 1 = c_1$.
   - **Result 1:** $c_1 = 2$.
2. **Test $f(x) = x$ (Linear):**
   - Exact: $\int_{-1}^{1} xdx = [\frac{1}{2}x^2]_{-1}^{1} = \frac{1}{2} - \frac{1}{2} = 0$.
   - Approximation: $c_1 \cdot f(x_1) = 2 \cdot x_1$.
   - **Result 2:** $2x_1 = 0 \implies x_1 = 0$.

Final $n = 1$ Rule: $\int_{-1}^{1} f(x)dx \approx 2f(0)$

Error Formula: $E(f) = \frac{f''(\xi)}{3}$

Usage Guide: This is identical to the Midpoint Rule. Use this for very fast approximations or when the function is expected to be nearly a straight line over the interval. It is the most efficient 1-point estimate possible.

2 Case $n = 2$: The Optimal Pair

Goal: Find c_1, c_2, x_1, x_2 such that the rule is exact for polynomials of degree 3.

Step-by-Step Derivation

We use the symmetry of the interval $[-1, 1]$ to assume $c_1 = c_2$ and $x_1 = -x_2$.

1. **Test $f(x) = 1$:**
   - Exact: $\int_{-1}^{1} 1dx = 2$.
   - Approximation: $c_1(1) + c_2(1) = 2c_1$.
   - **Solve:** $2c_1 = 2 \implies c_1 = 1, c_2 = 1$.
2. **Test $f(x) = x^2$:**
   - Exact: $\int_{-1}^{1} x^2dx = [\frac{1}{3}x^3]_{-1}^{1} = \frac{1}{3} - (-\frac{1}{3}) = \frac{2}{3}$.
   - Approximation: $1(x_1)^2 + 1(x_2)^2$. Since $x_1 = -x_2$, then $x_1^2 = x_2^2$.
   - **Solve:** $x_1^2 + x_1^2 = \frac{2}{3} \implies 2x_1^2 = \frac{2}{3} \implies x_1^2 = \frac{1}{3}$.
   - **Result:** $x_1 = -\frac{1}{\sqrt{3}} \approx -0.57735, \quad x_2 = \frac{1}{\sqrt{3}} \approx 0.57735$.

Final $n = 2$ Rule: $\int_{-1}^{1} f(x)dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$

Error Formula: $E(f) = \frac{f^{(4)}(\xi)}{135}$

Usage Guide: Use this when you have a function with unknown curvature. It is significantly more accurate than the Trapezoidal Rule while using the same number of function evaluations (2).

3 Case $n = 3$: The Precision Rule

Goal: Exactness for degree 5. We need 3 weights and 3 nodes.

Step-by-Step Derivation

Due to symmetry, the middle node x_2 must be 0. We set $x_1 = -x_3$ and $c_1 = c_3$.

1. **Test $f(x) = 1$:** $c_1 + c_2 + c_3 = 2c_1 + c_2 = 2$.
2. **Test $f(x) = x^2$:** $c_1x_1^2 + c_2(0)^2 + c_1(-x_1)^2 = 2c_1x_1^2 = \frac{2}{3}$.
3. **Test $f(x) = x^4$:** $c_1x_1^4 + c_2(0)^4 + c_1(-x_1)^4 = 2c_1x_1^4 = \frac{2}{5}$.
4. **Solve for x_1:** Divide the x^4 equation by the x^2 equation: $\frac{2c_1x_1^4}{2c_1x_1^2} = \frac{2/5}{2/3} \implies x_1^2 = \frac{3}{5}$. So, $x_1 = -\sqrt{0.6} \approx -0.77459$.

5. **Solve for c_1:** Substitute $x_1^2 = 3/5$ into $2c_1x_1^2 = 2/3 \implies 2c_1(3/5) = 2/3 \implies c_1 = \frac{5}{9}$.
6. **Solve for c_2:** $2(5/9) + c_2 = 2 \implies c_2 = 18/9 - 10/9 = 8/9$.

Final $n = 3$ Rule: $\int_{-1}^{1} f(x)dx \approx \frac{5}{9}f(-\sqrt{0.6}) + \frac{8}{9}f(0) + \frac{5}{9}f(\sqrt{0.6})$

Error Formula: $E(f) = \frac{f^{(6)}(\xi)}{15750}$
Usage Guide: This is the "gold standard" for manual calculations. It handles cubic and quartic curves with ease. Use this for engineering-level precision in smooth functions.

4 The "Roots Engine": Monic Legendre Polynomials

The optimal nodes x_i for any n-point rule are the roots of the n^{th} degree Monic Legendre Polynomial. These polynomials are orthogonal on $[-1, 1]$.

- $P_1(x) = x$ (Root: $x = 0$)
- $P_2(x) = x^2 - \frac{1}{3}$ (Roots: $x = \pm\sqrt{1/3} \approx \pm 0.57735$)
- $P_3(x) = x^3 - \frac{3}{5}x$ (Roots: $x = 0, \pm\sqrt{3/5} \approx \pm 0.77460$)
- $P_4(x) = x^4 - \frac{6}{7}x^2 + \frac{3}{35}$ (Roots: $\approx \pm 0.33998, \pm 0.86114$)
- $P_5(x) = x^5 - \frac{10}{9}x^3 + \frac{5}{21}x$ (Roots: $0, \pm 0.53847, \pm 0.90618$)
- $P_6(x) = x^6 - \frac{15}{11}x^4 + \frac{5}{11}x^2 - \frac{5}{231}$ (Roots: $\approx \pm 0.23862, \pm 0.66121, \pm 0.93247$)

5 Crucial: Interval Transformation

Gaussian Quadrature only works for $[-1, 1]$. To use it for $[a, b]$, use this change of variable:

$$x = \frac{(b-a)t + (b+a)}{2}, \quad dx = \frac{b-a}{2}dt$$

$$\int_a^b f(x)dx = \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{(b-a)t + (b+a)}{2}\right)dt$$

Usage Guide: Always apply this transformation first. It "stretches" or "shrinks" your interval to fit the standard $[-1, 1]$ template.

6 Worked Example: Applying $n = 2$ to $[1, 2]$

Problem: Approximate $\int_1^2 \frac{1}{x}dx$ using $n = 2$ Gaussian Quadrature.

1. **Transformation:** $a = 1, b = 2$. $x = \frac{2-1}{2}t + \frac{1+2}{2} = 0.5t + 1.5$. $dx = 0.5dt$. Integral becomes: $0.5\int_{-1}^{1} \frac{1}{0.5t+1.5}dt$.
2. **Evaluations ($n = 2$ nodes):** $t_1 = -0.57735 \implies g(t_1) = \frac{1}{0.5(-0.57735)+1.5} \approx 0.82554$. $t_2 = 0.57735 \implies g(t_2) = \frac{1}{0.5(0.57735)+1.5} \approx 0.55907$.
3. **Summing with weights ($c = 1$):** Result $= 0.5 \times [1(0.82554) + 1(0.55907)] = 0.5 \times [1.38461] = \mathbf{0.69231}$.
4. **Verification:** Exact value is $\ln(2) \approx 0.69315$. The error is only 0.00084.

7 Higher-Order Formulas ($n = 4, 5, 6$)

For high-precision needs, use the following tabulated nodes (x_i) and weights (c_i) for the interval $[-1, 1]$.

For $n = 4$ (Exact for degree 7)

$\int_{-1}^{1} f(x)dx \approx 0.347855f(-0.861136) + 0.652145f(-0.339981) + 0.652145f(0.339981) + 0.347855f(0.861136)$

For $n = 5$ (Exact for degree 9)

$\int_{-1}^{1} f(x)dx \approx 0.236927f(-0.906180) + 0.478629f(-0.538469) + 0.568889f(0) + 0.478629f(0.538469) + 0.236927f(0.906180)$

For $n = 6$ (Exact for degree 11)

$\int_{-1}^{1} f(x)dx \approx 0.171324[f(-0.932470) + f(0.932470)] + 0.360762[f(-0.661209) + f(0.661209)] + 0.467914[f(-0.238619) + f(0.238619)]$